

Adaptive Control & RL



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Adaptive Control & RL

Outline

- Introduction
- Stability of Dynamical Systems
- Adaptive Control
- Robust Neuro-Adaptive Control (Stability & Design)
- Reinforcement Learning & Neuro-Adaptive Control

Adaptive Control & RL

To study

- Slides.
- Technical Report.
- Text books:
 - A. J.-J. E. Slotine, W. Li, *“Applied Nonlinear Control”*
 - B. H. K., Khalil, *“Nonlinear Systems”*
 - C. J. T. Spooner, M. Maggiore, R. Ordóñez, K. M. Passino, *“Stable Adaptive Control and Estimation for Nonlinear Systems: Neural and Fuzzy Approximator Techniques”*
 - D. J. A. Farrell & M. M. Polycarpou, *“Adaptive Approximation Based Control-Unifying Neural, Fuzzy and Traditional Adaptive Approximation Approaches”*

For more

<https://www.incontrolpodcast.com/1632769/episodes/12037544-ep8-anuradha-annaswamy-adaptive-control-from-the-brave-era-to-reinforcement-learning-and-back>

<https://tinyurl.com/yck89red> <https://tinyurl.com/yc27rsyd>

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Evaluation

- **Mandatory Homeworks** (3/10 points)
[During the semester]
- **Mandatory Project** (4/10 points)
[End of semester]
- **Mandatory Lab** (3/10 points)
[Middle of semester]



<https://janismac.github.io/ControlChallenges/>

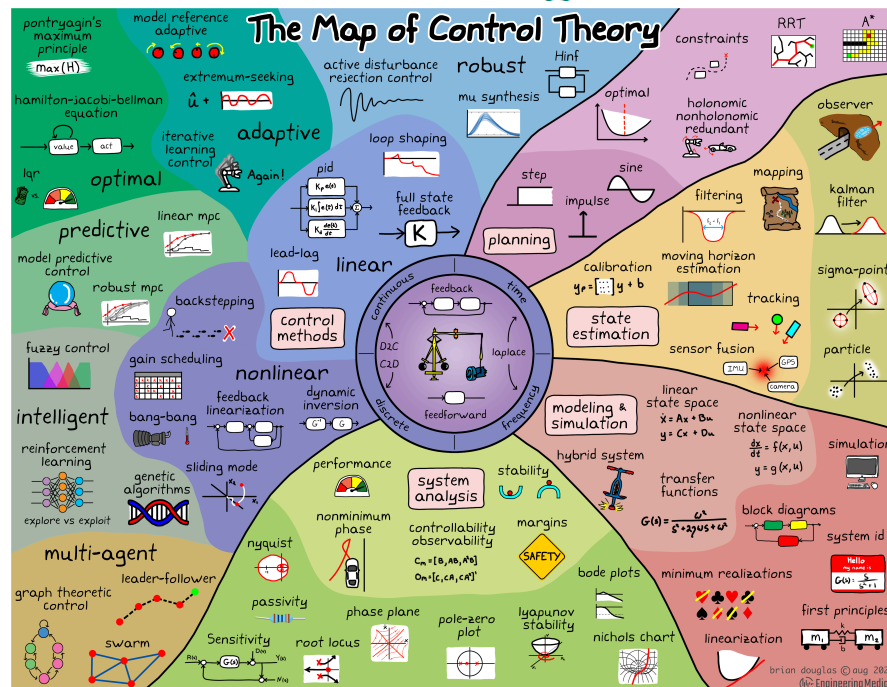
```
var delta_t = 0.02; // The simulation time step
position_error_integral += delta_t * block.x;
monitor('var',[block.T,block.x,block.dx]);
const data = document.querySelector('#variableInfo').innerText;
console.log(data);
return -3*block.x -5*block.dx -2*position_error_integral;
```

- Modeling
- Identification
- Adaptive control design
- Implementation

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Course description

- This course introduces the general principles of adaptive control and learning



Adaptive Control & RL

Why adaptive control?

- Systems to be controlled have parameter uncertainty
- System dynamics experience unpredictable parameter variations as the control operation goes on

Adaptive Control & RL

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- **Examples**
 - Robot manipulation



Adaptive Control & RL

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Adaptive Control & RL

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- **Examples**
 - Robot manipulation
 - Ship steering
 - Aircraft control



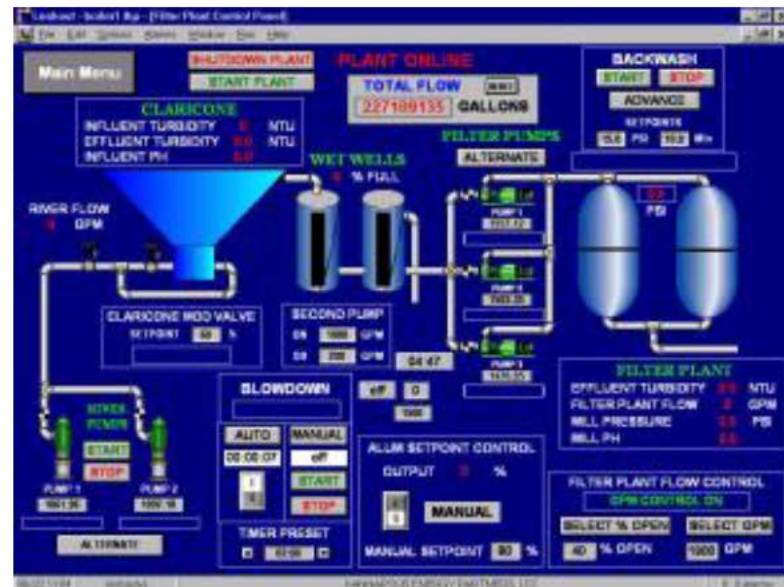
Adaptive Control & RL

Why adaptive control?

- Systems to be controlled have parameter uncertainty
- System dynamics experience unpredictable parameter variations as the control operation goes on

- **Examples**

- Robot manipulation
- Ship steering
- Aircraft control
- Process control



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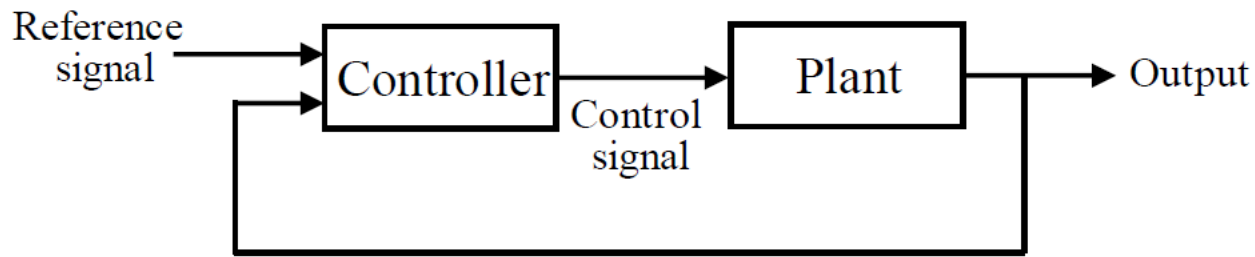
What is adaptive control?

- Definition
 - A form of control in which the control parameters are automatically adjusted as conditions change so as to optimize performance [cited from Oxford English Dictionary]

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What is adaptive control?

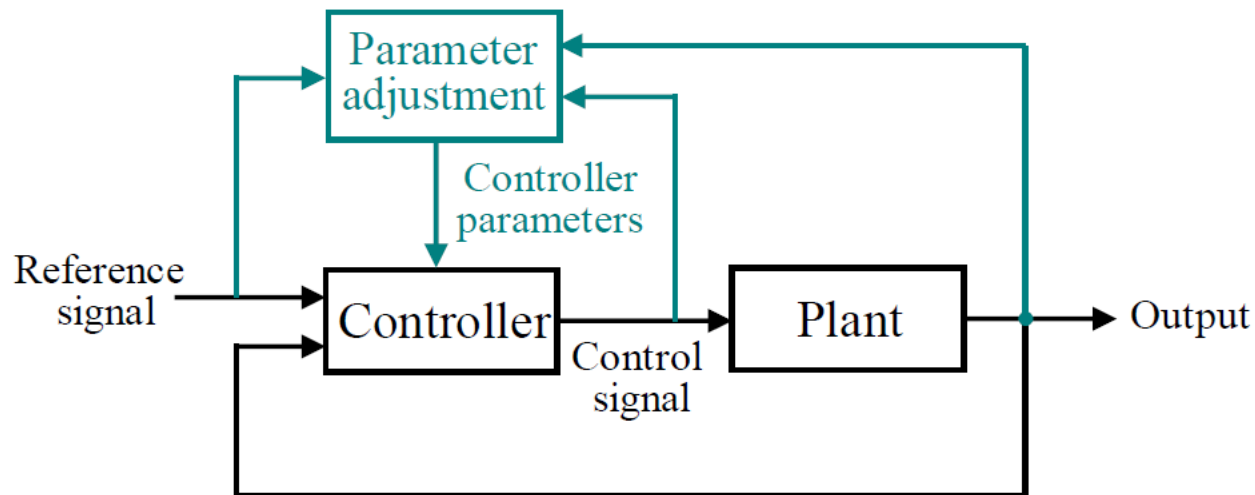
- Definition
- Difference between adaptive control and ordinary feedback control



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What is adaptive control?

- Definition
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What is adaptive control?

- Definition
- Difference between adaptive control and ordinary feedback control
- **Difference between adaptive control and robust control**
 - Robust control can also be used to deal with parameter uncertainty
 - Adaptive control is superior to robust control in dealing with uncertainties in constant or slowly-varying parameters
 - Robust control has some desirable features which adaptive control does not have, such as its ability to deal with quickly varying parameters and unmodeled dynamics

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What is adaptive control?

- Definition
- Difference between adaptive control and ordinary feedback control
- Difference between adaptive control and robust control
- A brief history
 - In 1950s, connection with design of autopilots for high-performance aircraft



F101A (VooDoo)



Adaptive Control & RL

What is adaptive control?

- Definition
- Difference between adaptive control and ordinary feedback control
- Difference between adaptive control and robust control
- A brief history
 - In 1950s, connection with design of autopilots for high-performance aircraft
 - During that time, the work on adaptive control was characterized by a “lot of enthusiasm, bad hardware and nonexistent theory” [Astrom, 1983]. The interest in adaptive control soon diminished due to the lack of insights and stability proofs coupled by the crash of a test flight

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What is adaptive control?

- Definition
- Difference between adaptive control and ordinary feedback control
- Difference between adaptive control and robust control
- **A brief history**
 - In 1950s, connection with design of autopilots for high-performance aircraft
 - In late 1960s, advances in control theory contributing to the development of adaptive control (e.g., state-space description, Lyapunov's stability theory, and dynamic programming)

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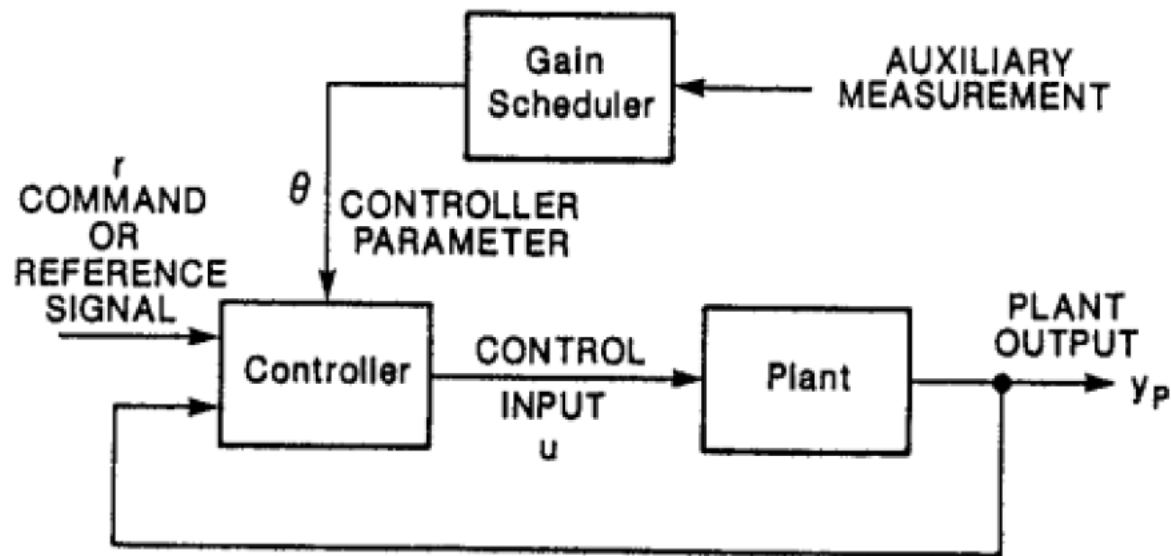
What is adaptive control?

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- **A brief history**
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 - In 1960s, advances in control theory contributing to the development of adaptive control (e.g., state-space description, stability theory, and dynamic programming)
 - In 1970s and early 1980s, proofs for stability of adaptive systems
 - In late 1980s and early 1990s, new insights into robustness of adaptive controllers
 - Recently, strong relations to ideas on **learning**

Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling



Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling
 - Advantages of gain scheduling
 - Parameters can be changed quickly in response to changes in the plant dynamics. It is convenient especially if the plant dynamics depend in a well-known fashion on a relatively few easily measurable variables

Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling
 - Advantages of gain scheduling
 - Disadvantages of gain scheduling
 - It is an open-loop adaptation scheme, with no real “learning” or intelligence
 - The extent of design required for its implementation can be enormous (example: flight control system of CH-47 helicopter)

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Approaches to adaptive control

- Gain scheduling

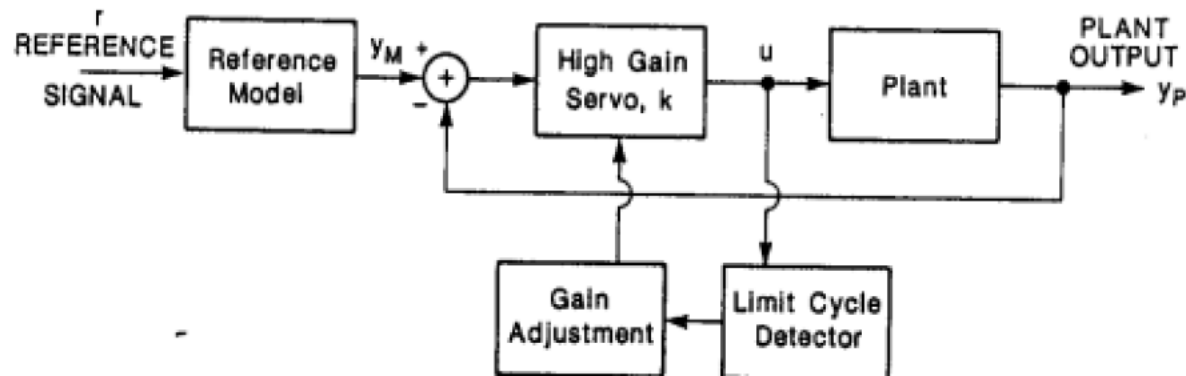
CH-47
(Chinook)



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Approaches to adaptive control

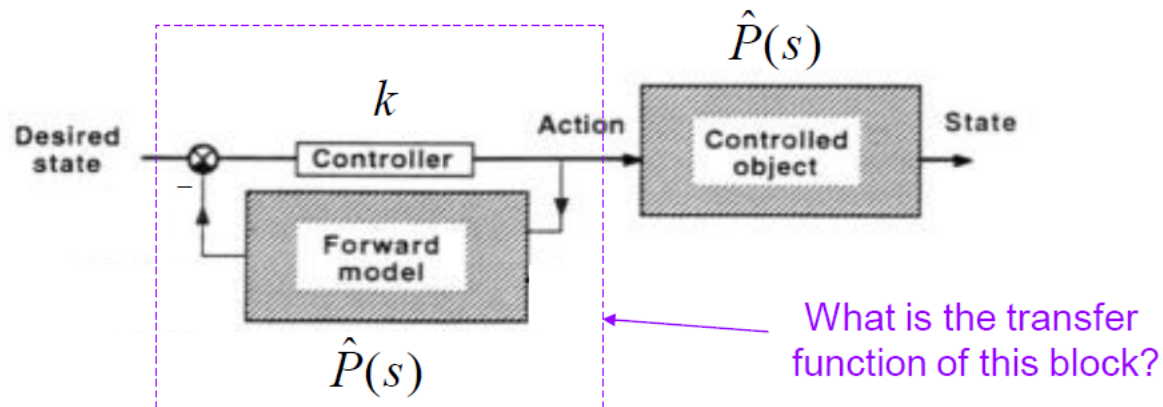
- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme



Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling
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Using high-gain and a plant model to implement an inverse model of the plant

Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme
 - Advantages: intuitive and simple—only one parameter is updated

Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme
 - Advantages
 - Disadvantages:
 - (1) Oscillations are constantly present in the system
 - (2) Noise in the frequency band of the limit cycle detector causes the gain to decrease well below the critical value
 - (3) Reference inputs may cause saturation due to the high-gain
 - (4) Saturation may mask limit cycle oscillations, allowing the gain to increase above the critical value, and leading to instability

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Approaches to adaptive control

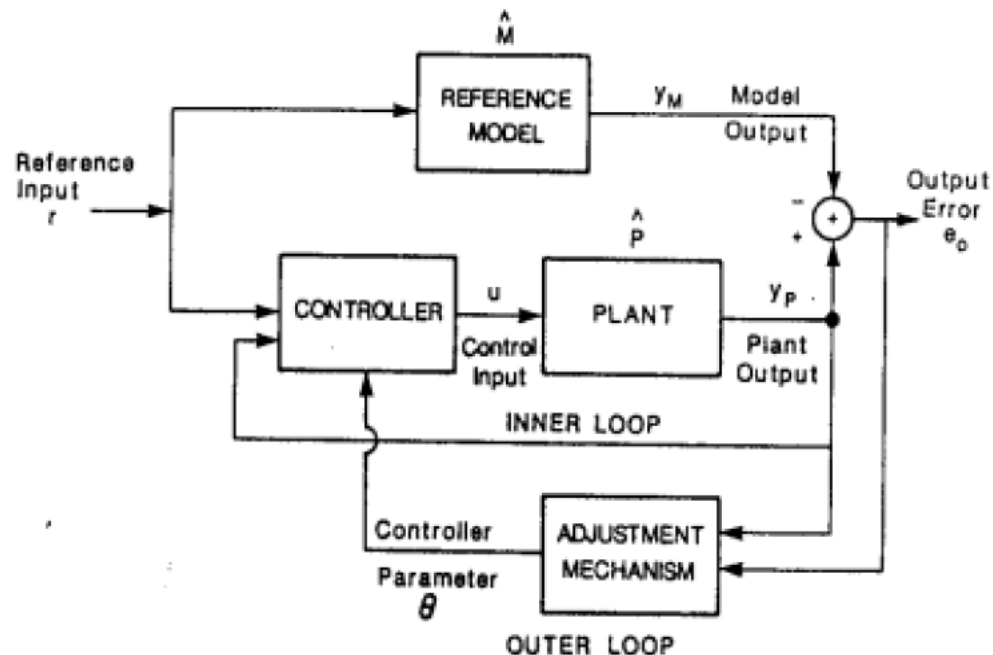
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- Model reference adaptive systems
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Adaptive Control & RL

Approaches to adaptive control

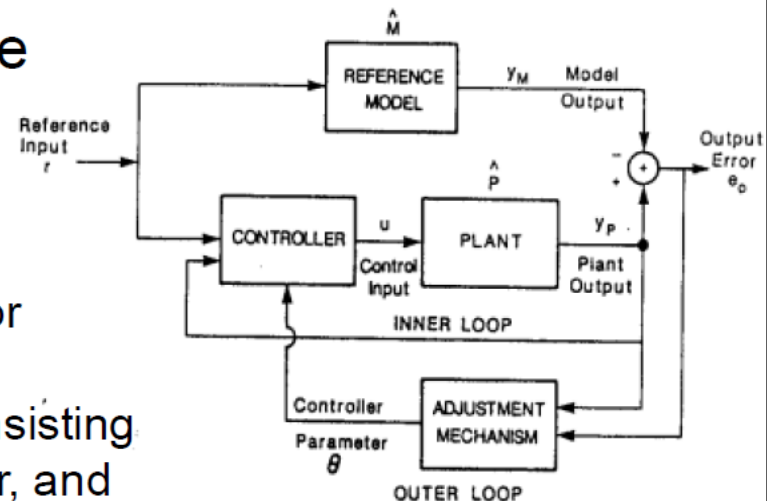
- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme
 - Parallel scheme



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Approaches to adaptive control

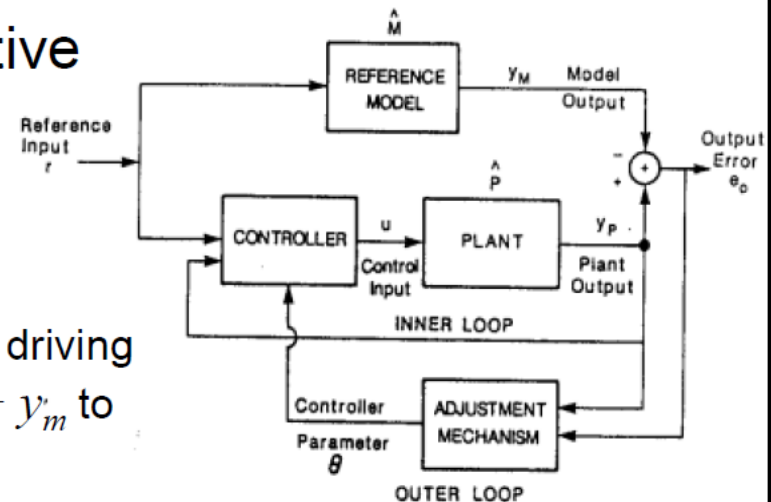
- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme
 - Parallel scheme
 - Two loops: (1) an **inner** or **regulator loop** that is an ordinary control loop consisting of the plant and regulator, and (2) an **outer** or **adaptation loop** that adjusts the parameters of the regulator in such a way as to drive the error between the model output and plant output to zero



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Approaches to adaptive control

- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme
 - Parallel scheme
 - Two loops
 - Adjustment mechanism: driving the output error $e_0 = y_p - y_m$ to zero



Gradient update

$$\begin{aligned} \frac{d\theta}{dt} &= -\underset{\substack{\text{Adaptation gain}}}{g} \frac{\partial}{\partial \theta} (e_0^2(\theta)) \\ &= -2ge_0(\theta) \frac{\partial}{\partial \theta} (e_0(\theta)) = -2ge_0(\theta) \frac{\partial}{\partial \theta} (y_p(\theta)) \end{aligned}$$

Sensitivity function

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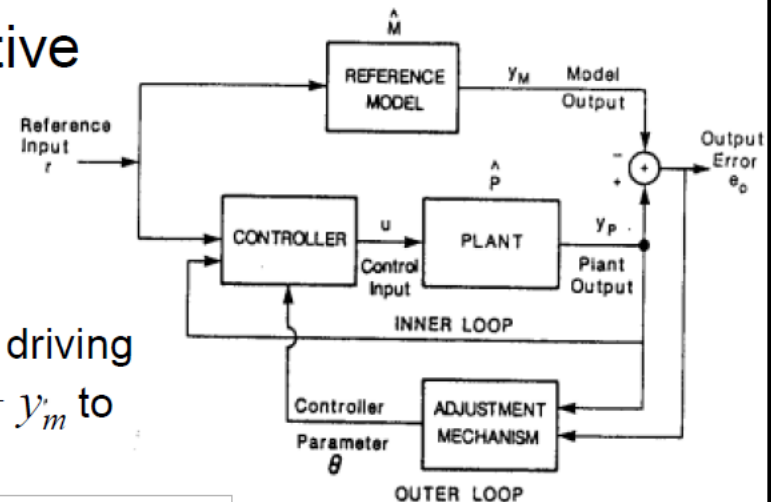
Approaches to adaptive control

- Gain scheduling
- Model reference adaptive systems
 - Series high-gain scheme
 - Parallel scheme
 - Two loops
 - Adjustment mechanism: driving the output error $e_0 = y_p - y_m$ to zero

Gradient update

$$\frac{d\theta}{dt} = -2ge_0(\theta) \frac{\partial}{\partial \theta} (y_p(\theta))$$

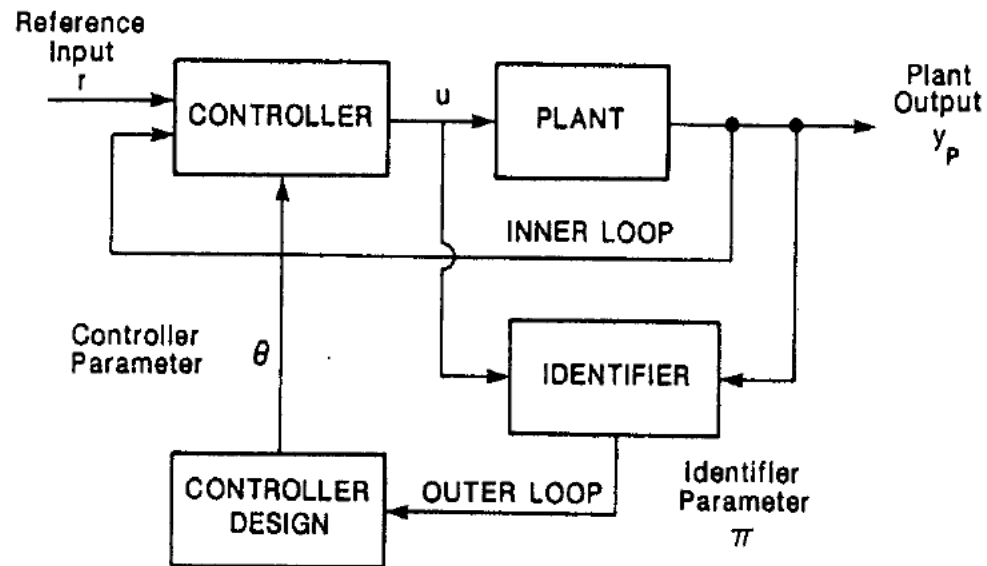
M.I.T. rule: an implementation of the gradient update by replacing the unknown parameters in $\partial y_p(\theta)/\partial \theta$ by their estimates at time t



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Approaches to adaptive control

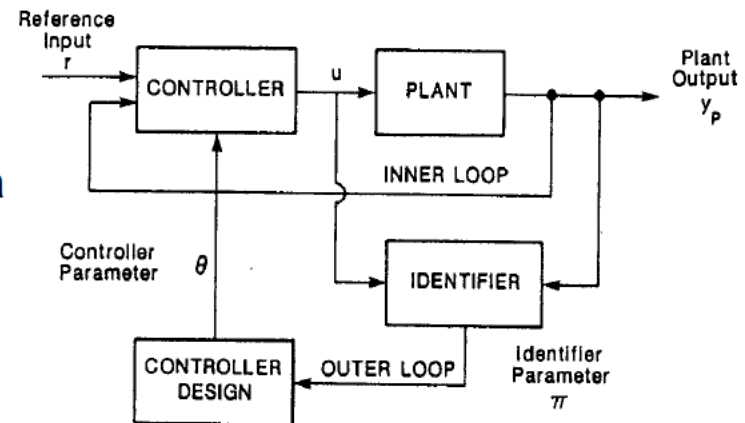
- Gain scheduling
- Model reference adaptive systems
- Self tuning regulators
 - Certainty equivalence principle



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Approaches to adaptive control

- Gain scheduling
- Model reference adaptive systems
- **Self tuning regulators**
 - Certainty equivalence principle
 - Two loops
 - Inner loop consisting of a conventional controller, but with varying parameters
 - Outer loop consisting of an identifier and design box which adjust these controller parameters



Adaptive Control & RL

Approaches to adaptive control

- Gain scheduling
- Model reference adaptive systems
- Self tuning regulators
 - Certainty equivalence principle
 - Two loops
 - Direct and indirect adaptive control

